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# Work-life balance and well-being of graduate students

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**ABSTRACT**
This exploratory study examines issues related to work-life balance and well-being of a diverse population of graduate students, including master's and doctoral students, full-time and part-time graduate students, on-campus and on-line students, and from multiple disciplines. Using data from graduate students at a large, public university on the mid-Atlantic coast, our results provide a broad picture of work-life balance, three components of well-being – quality of life, physical health, and mental health – and factors such as program climate, sense of well-being, stressors, and sources of support. We discuss how social cognitive, structuration, and border theories can be applied to study and address issues of work-life balance and well-being in more depth.

**KEYWORDS**
Graduate students; work-life balance; mental health; well-being; quality of life

## Introduction

The attention to a population's well-being has been increasing over the decades, and well-being has been recognized as being associated with physical, psychological, and mental health, and lifestyle choices (Diener, 2006). Well-being can be determined by a combination of life circumstances and experiences that include personal and professional roles, and social and economic status, among others. Well-being is multidimensional, and various factors can affect well-being. Work-life balance is one key issue related to well-being, and the connection between the two has been a focus for university faculty as the demands of academic jobs can reduce well-being. However, as noted by Evans et al. (2018), much less is known about work-life balance of graduate students despite rising tuition costs, growing student debt, and more students working full-time while completing graduate work. In addition, recent studies have demonstrated mental health concerns of doctoral students (Barreira et al., 2018; Cornwall et al., 2019; Evans et al., 2018; Janta et al., 2014; Waight & Giordano, 2018). For example, Barreira et al. (2018) found that 18% of economics doctoral students experienced moderate or severe symptoms of depression and anxiety, and about one in ten reported having suicidal thoughts. However, more research is needed to better understand mental health issues among the broader graduate student population, especially since this population has become more diverse (Brus, 2006). The majority of studies have focused on full-time doctoral students, resulting in less-informed understanding of work-life balance and well-being issues of part-time or nontraditional graduate students who take courses on-campus or at a distance.

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Our study adds to the current literature by exploring work-life balance and well-being of a diverse population of graduate students, such as part-time and full-time students, on-campus and on-line learners, and master's and doctoral students; these students extend across a broad range of disciplines. We ask the question: *how is work-life balance connected to the well-being of graduate students?* Specifically, we seek to understand the graduate student experience in relation to both academic and personal components of graduate students' lives and the implications for their well-being. As such, we examine related factors – such as program climate, sense of belonging, sources of stress, and support networks – that are identified in the literature as important for graduate students' work-life balance and well-being.

## Graduate student work-life balance, well-being, and related factors

Imbalance between multiple and competing roles and responsibilities of graduate students may cause stress. Well-being, which incorporates quality of life and physical and mental health, can be impacted by this stress. Well-being can also be affected by such factors as program climate, sense of belonging to the university community, and sources of support.

### *Work-life balance*

Work-life balance in higher education has primarily been studied in the context of faculty (Brus, 2006; Marshall et al., 2012; Moyer et al., 1999; Stimpson & Filer, 2012). Fewer studies have focused on work-life balance of graduate students. The studies that have done so tend to focus on doctoral students and do not consider the diversity of graduate students in the U.S. For graduate students, work-life balance can be considered in terms of the attainment, enjoyment, and satisfaction with the amount and quality of time committed to different facets of school, work, and life (Martinez et al., 2013). "Work" includes course work, research requirements, career and professional development activities, and other academic-related activities that graduate students engage in. "Life," on the other hand, encompasses the non-academic, personal components of graduate student lives such as family and social life, and non-school activities associated with employment.

Graduate students have multiple and often competing roles and responsibilities, each demanding time and attention (Brus, 2006; Haynes et al., 2012; Martinez et al., 2013; Moyer et al., 1999; Offstein et al., 2004; Stimpson & Filer, 2012). These roles include being students, graduate assistants, full-time employees, primary breadwinners, parents, spouses, caretakers, and friends. Meeting the demands of multiple roles can affect the quality of life, as graduate students must make tradeoffs, such as in terms of financial security, time with family and friends, and time for self. Brus (2006, p. 31) recognized that doctoral students "struggle to balance their academic pursuits with their personal lives and responsibilities," and these struggles are sources of stress (Brus, 2006; Cornwall et al., 2019; Martinez et al., 2013; Offstein et al., 2004). Offstein et al. (2004) found that doctoral students described their workload as time- and effort-intensive, and their lives as stressful, pressed, and lacking breathing space.

The literature on doctoral student work-life balance has found gender imbalances with women reporting less balance (Brus, 2006; Haynes et al., 2012; Kulp, 2020; Martinez et al., 2013; Moyer et al., 1999; Stimpson & Filer, 2012), primarily due to challenges of family commitments and caregiving roles (Brus, 2006; Kulp, 2020; Stimpson & Filer, 2012). Doctoral students who are mothers, for example, "face unique identity, role, work-family and organizational conflicts" that can affect their graduate school experience (Kulp, 2020, p. 3).

Balancing work-life requires constant decision-making and weighing of trade-offs (Brus, 2006; Cornwall et al., 2019; Haynes et al., 2012; Martinez et al., 2013; Offstein et al., 2004). Not surprisingly, time management is one of the most commonly reported difficulties associated with balancing multiple responsibilities (Barreira et al., 2018; Brus, 2006; Cornwall et al., 2019; Martinez et al., 2013; Offstein et al., 2004). As Martinez et al. (2013) found, a key struggle for full-time doctoral students was purposefully managing their time, priorities, and roles and responsibilities. Time management is a key element of work-life balance for graduate students, as time devoted to work means less time allowed for other life-related activities, which can then impact well-being.

### *Well-being*

As a broad concept, well-being "encapsulates an individual's quality of life, happiness, satisfaction with life and experience of good mental and physical health" (Noble & McGrath, 2012, p. 32). Well-being is multi-dimensional, encompasses individual perceptions of one's experiences, and has health and social or subjective perspectives (Haynes et al., 2012; Helliwell & Putnam, 2004; Juniper et al., 2012; Schmidt & Hansson, 2018). For this study we define well-being as multi-dimensional with three components: quality of life, physical health, and mental health.

Subjective well-being or quality of life is defined as the "individual's perception of their position in life, in the context of the culture and value systems in which they live, and in relation to their goals, expectations, standards, and concerns" (WHOQOL Group, 1995, p. 1404). According to a 2014 report by the University of California (UC) Berkeley Graduate Assembly, quality of life varies across different groups of graduate students and is sensitive to different components of "work" and "life" and the balance of both. For instance, among graduate students, women are consistently more likely to have mental health issues than men (Barreira et al., 2018; Evans et al., 2018).

The World Health Organization explicitly links health and well-being, and in the context of our study, the mental health aspect is particularly relevant as "stress is at the core of the graduate student experience and is amplified by conflicting demands and internal conflict" (Offstein et al., 2004, p. 396). Studies show that graduate students often suffer from anxiety, depression, and other mental health issues (Barreira et al., 2018; Evans et al., 2018; Levecque et al., 2017; Waight & Giordano, 2018). Graduate students were much more likely to experience depression and anxiety relative to the general population (Evans et al., 2018). For example, a study at UC Berkeley found that 43% to 46% of graduate students were depressed (UC Berkeley Graduate Assembly, 2014). The UC Berkeley Graduate Assembly (2014) report also found that about 44% of students they surveyed reported being physically sick or ill during the semester. Hyun et al. (2006) found that almost half of all graduate students reported experiencing emotional- or stress-related problems over the past year and more than half reported knowing a fellow graduate student who had experienced such problems.

Stress resulting from lack of or trying to achieve work-life balance can have physical and mental health effects. For example, more than half of Master's and doctoral students in the study by Evans et al. (2018) who experienced moderate to severe anxiety also indicated they did not have good work-life balance. As noted by Juniper et al. (2012), there is a need for more studies on graduate students and broader focus on their well-being. In this study we approach well-being in a wholistic way, incorporating both a quality of life or subjective well-being dimension along with the health dimension consisting of both mental and physical health.

### *Factors affecting work-life balance and well-being*

Research shows that various factors can play a role in graduate student well-being. For example, the UC Berkeley Graduate Assembly (2014) study identified ten important predictors of graduate student satisfaction with life and experience with depression, such as academic engagement, social support, financial confidence, relationship with advisors, feeling valued and included, and academic progress and preparation. For doctoral students, Juniper et al. (2012) identified additional relevant factors, such as the research experience, social activities, and professional and career development, and Cornwall et al. (2019) found several stress-related themes, including sense of belonging, time pressures, and social isolation.

Social support is a consistent theme throughout the literature on graduate education, particularly studies addressing mental health, work-life balance, and student success. Graduate students rely on various support structures as they strive to achieve well-being and a balance between work and life (Martinez et al., 2013), including family, friends, coworkers, faculty advisors and mentors, and peers and other graduate students. This support ranges from emotional to professional, such as course advising or mentoring, to practical, such as financial resources or assistance with tasks (Barreira et al., 2018; Martinez et al., 2013; Offstein et al., 2004). Graduate students' support structure can provide venues to relieve stress or reduce social isolation. Barreira et al. (2018) found that "supportive and collaborative classmates, people who understand the PhD experience, and others who can be trusted to have the student's best interest in mind appear to be extremely valuable tools for mitigating shocks to mental health" (p. 23).

While faculty and advisors may be sources of support for graduate students (Bieber & Worley, 2006; Jairam & Kahl, 2012), the relationship can be a central concern of many students and may not be supportive of work-life balance and well-being (Barreira et al., 2018; Evans et al., 2018; Lee, 2008; Al Makhamreh & Stockley, 2019). Barreira et al. (2018) found that among economics doctoral students, most had good, helpful relationships with their faculty advisors, but many did not receive substantial support from their advisors. For example, 27% of women and 35% of men perceived that their advisors do not care about them, and 36% of students reported not having had, in the previous two months, any informal conversations with faculty about how they were doing academically or professionally (Barreira et al., 2018). Evans et al. (2018) found similar results, but also recognized that "strong supportive and positive mentoring relationships … correlate significantly with less anxiety and depression" (p. 283).

Graduate students may also not see advisors filling a social support role. For example, graduate students may see their connection to faculty as more academic than social (Martinez et al., 2013). This is consistent with findings by Barreira et al. (2018) that fewer than 10% of economics doctoral students felt they can be very honest with their advisors about personal issues.

## Some insights from theory

While this is study is an exploration of how work-life balance connects to the well-being of graduate students, we can draw from theories to focus and situate our exploratory analysis of the graduate student experience in relation to both academic and personal components of graduate students' lives and the implications for their well-being. We will briefly introduce three theories – social cognitive theory, structuration theory, and border theory – that can help us better understand issues related to graduate students work-life balance including institutional elements that contribute to work-life imbalance and support structures and communication mechanisms to alleviate work-life balance challenges. These theories can also guide next steps beyond our exploratory findings. For example, the theories can inform development of policies and programs to help students bridge roles and responsibilities across multiple domains or can be applied to consider how future practices can be more responsive to work-life balance and well-being issues of graduate students. These implications for policy and practice will be discussed in the Conclusions section.

The first theory, social cognitive theory, centers around the concept of emergent interactive agency in that individuals cognitively, affectively, and behaviorally interact with their environments in a system of "triadic reciprocal causation" (Bandura, 1986b, 1989). In this system of reciprocal causality, internal personal factors interact with behavioral patterns and environmental factors, all working with bidirectional influence. Within this theoretical framework, human functioning is analyzed as socially interdependent, richly contextualized, and conditionally orchestrated within the dynamics of various societal subsystems and their complex interplay. Two key elements play a role in how people respond in any situation or context: the modifiability of the environment and the believed personal self-efficacy of individuals to change their environment through their actions (Bandura, 1989). People set goals for themselves, anticipate the likely consequences of their prospective actions given the environment within which they are embedded, and select and create courses of action likely to produce desired outcomes and avoid detrimental ones (Bandura, 2001).

Perceived self-efficacy is a critical element to understanding how graduate students are at the center of the issue regarding their work-life balance and well-being. Human attainment – such as achieving work-life balance – and positive well-being require an optimistic sense of self-efficacy (Bandura, 1986a). The stronger their perceived self-efficacy, the higher the goals people set for themselves, the more positively they visualize success scenarios, and the firmer their commitment to achieving these goals. A high sense of efficacy fosters cognitive constructions of effective actions, and cognitive reiteration of efficacious courses of action strengthens self-perceptions of efficacy (Bandura & Adams, 1977). Furthermore, how graduate students see themselves within the university structure and their perceived self-efficacy to advocate for and pursue work-life balance, then, has implications for their stress, anxiety, work-life balance, and, ultimately, well-being. This self-efficacy can be negatively impacted by other issues that graduate students concurrently face, such as lack of financial resources and less supportive climates within academic programs.

Personal efficacy also affects how much stress and depression people experience in challenging situations. Stress, a key component of work-life balance and well-being in the extant literature, results from the mismatch between perceived coping capabilities and potentially aversive aspects of the environment; those who believe they cannot manage potential challenges experience high levels of stress and anxiety (Bandura, 1998; Folkman &

Lazarus, 1984). Furthermore, socio-structural factors operate through psychological mechanisms (such as self-efficacy) to produce behavioral effects. For example, economic conditions, socioeconomic status, and family structures can indirectly affect behavior by impacting a person's aspirations, sense of efficacy, and personal standards (Bandura, 1993).

The second theory, structuration theory, emphasizes how structures may affect self-efficacy and drive action. In structuration theory people are autonomous and knowledgeable actors who know a great deal about the conditions and consequences of what they do (Giddens, 1984). Actors within systems or organizations engage in actions as a continuous flow of interventions rather than as discrete acts. These actions are purposive and monitored by actors who continually reflect on what they are doing, how others react to what they do, and the circumstances in which they find themselves. They use interpretive schemes to constitute and communicate meaning and act with intended and unintended consequences.

Structuration theory is useful for understanding how people experience work-life issues and how these experiences reinforce organizational practices and views (Hoffman & Cowan, 2010). It offers a lens for understanding the interactions and activities that shape graduate students' experiences with balancing work and life. Two concepts are important for examining work-life balance in an organizational context: the nature of structures, and the duality of structure (Hoffman & Cowan, 2010).

First, structures consist of rules and resources that affect behavioral choices (Sewell, 1992). Rules, whether official or learned through experiences, are guidelines that shape people's actions. For graduate students, official rules are usually outlined in graduate program handbooks or graduate catalogs. However, unofficial or learned rules can be contextual and nuanced in terms of specific interactions with individual faculty and advisors or be broadly applied across programs, colleges, or the university. Resources, whether material (books, budgets, computers) or nonmaterial (traditions, relationships), are available for use by the social actor to facilitate activities (McPhee et al., 2013). For graduate students negotiating work and life, rules might include official policy in the graduate program handbook or course policy in the syllabus. Work-related resources might include research supplies, a positive relationship with academic support staff, or a supportive social climate.

Second, structuration theory posits that rules and resources employed by actors are reproduced in routinized social practices that perpetuate and govern the continuity or transformation of the structures themselves (Canary & Tarin, 2017). The duality of structures concept refers to the mutually constitutive and implicative relationships between agents and structures. When people act in ways that abide by rules and use resources to carry out those actions they reinforce existing structures by enacting them and routinizing them in practice (Canary & Tarin, 2017; Poole & McPhee, 2005; Sewell, 1992). Structuration theory highlights how patterns of behaviors create a guiding system of future action, where past behavioral patterns reinforce consistent behaviors and constrain inconsistent ones.

Graduate education involves relationships between various actors – graduate students, faculty, advisors, program staff, administrators, etc. – within which actors must explain and justify their actions to others. In this sense, program climate is a key contributor to work-life balance and related well-being issues. Graduate students also face expectations that are manifested in rules and patterns followed in organizational life within the academic program and the university, and the fulfillment of such expectations can come with resources (and lack of fulfillment may come with sanctions). When graduate students (1) act in ways that create work-life imbalances by, for example, conforming to implicit rules that school must always come first and that a social life is unimportant, and use work-related resources to prioritize work activities, and then (2) are rewarded by faculty or advisors for taking such actions, the recursive structure reinforces that work-life imbalance is a facet of being a graduate student and that students should continue to behave in ways that emphasize work over life. One way to address work-life challenges of graduate students is to understand the rules and resources at play when graduate students interact with their peers, faculty, advisors, department chairs, and others on issues related to work-life balance, and how those interactions reinforce behaviors detrimental to work-life balance and well-being. The intellectual and social climate of academic programs can shape perceptions of these rules and resources and whether behaviors consistent with work-life balance are reinforced.

Finally, work/family border theory presents a useful descriptive framework for understanding graduate student work-life balance and the respective actions that can be taken by graduate students themselves to manage work-life, and by supporting actors – faculty, advisors, department chairs, peers, etc. – to facilitate graduate students' management of the boundaries between work and life. The theory recognizes that work and family (or life in the context of our study) are interconnected and that people are border-crossers who must regularly traverse the borders between work and life domains.

Work/family border theory can inform understanding of when and where work-life conflict may arise (Clark, 2000; Schieman et al., 2009). It recognizes factors that contribute to how different groups of graduate students manage work-life balance. The theory recognizes that people vary in influence, or the degree to which they have power to mold each domain and traverse its boundaries, and the degree to which membership within a particular domain is salient and congruent with their identity (Clark, 2000). Research suggests that women prefer to keep work and family domains as either distinct or integrated, and those preferences will affect the attributes of the borders between the domains and their movement across them (Nippert-Eng, 2008). Expectations for border crossing, then, may vary for graduate students of different genders. The extent to which domains are different may also cause imbalance. When domains are significantly different, the area of blending between domains can be "places where border-crossers awkwardly juggle conflicting demands and conflict arises" (Clark, 2000, p. 754), but strong borders can facilitate work-life balance. Full-time graduate students may struggle more than part-time students in terms of keeping their work and family/life domains separate.

The theory suggests that individuals actively create borders between work and life (Nippert-Eng, 2008); borders are demarcation lines between domains, defining the point at which domain-relevant behaviors begin or end (Clark, 2000). For graduate students, work-life may be separated by physical borders (that define where domain-relevant behaviors take place), temporal borders (that define when work is done and life responsibility is assumed), and psychological borders (that dictate appropriate thinking patterns, behavior patterns, and emotions for the respective domains). Understanding the characteristics of borders between work and life for different types of students is important for understanding how they navigate the borders and manage work-life conflict. Border theory provides several avenues for considering policies and practices that encourage graduate student work-life balance and support graduate student well-being.

Issues related to graduate student health and well-being have come to the forefront of discussion both within and outside of the academy. To answer our research question – *How is work-life balance connected to the well-being of graduate students?* – this study examines various factors identified in the literature as important for understanding work-life balance and well-being of graduate students, including program climate, sense of belonging to the university community, sources of stress, and sources of support.

## Methods and analysis

Unlike the majority of studies that focus on full-time doctoral students, our study provides a more encompassing answer to the question of how work-life balance is connected to the well-being of a wider graduate student population. We do so by using a sample of graduate students from one university that is inclusive of the many different types of graduate students at U.S. universities, including master's and doctoral students, full-time and part-time students, on-campus and on-line students, and across different disciplines. However, our analysis does not differentiate between domestic (U.S.) or international students.[1] Our methodology is consistent with other studies that have utilized samples from a single U.S. institution (UC Berkeley Graduate Assembly, 2014; Martinez et al., 2013; Offstein et al., 2004). While we recognize the limitations of using a single organization for our study, the use of this single university as a study site allowed access to a diverse population of graduate students that is more representative of graduate students across the country.

Our analysis uses data from a survey of graduate students at a large, public university in the U.S. The online survey was conducted in April and May 2018 – from the middle of the semester through a week after final exams – as part of a broader effort to better understand the graduate student experience at the university. E-mail invitations to participate in the survey were sent to all graduate students at the university, and participants were assured that their responses were anonymous. A total of 343 students enrolled in graduate degree programs at the University completed the survey, representing 9.2% of all degree-seeking graduate students at the university.

The instrument was modeled after surveys used by other U.S. universities to assess their graduate students' experiences.[2] Questions about program climate, mental health challenges, and sources of support came from the MIT Enrolled Graduate Student Survey. The question regarding sense of belonging came from the Temple University Graduate Student Survey and the University of Washington Annual Graduate Student Survey. The question regarding stress incorporated sources of stress asked about in multiple surveys (MIT Enrolled Graduate Student Survey, University of Washington Annual Graduate Student Survey, Temple University Graduate Student Survey, and UC Graduate Student Well-being Survey). In recognizing the diversity of graduate students, our survey questions and responses were designed to be sufficiently broad to capture the diversity of issues facing graduate students.

It is important to note, however, that the survey was designed primarily for the institutional purpose of assessing the graduate student experience, with the research purpose being secondary. The survey questions were grounded in practical applications of how graduate student well-being can be addressed, rather than being grounded in the research literature. As such, the construction of the survey questions emphasized ease of identifying key issues rather than for conducting multivariate analysis of relationships between variables. This duality of purposes is a limitation of our study.

### Background of the university

The university is an urban university located on the Mid-Atlantic coast of the U.S. in a metropolitan area that is diverse in terms of race, ethnicity, and income. The university has more than 90 undergraduate programs and more than 60 graduate programs at the master's and doctoral levels. The university has an extensive on-line education program with more than 100 programs (total) available through that modality.

In the semester of data collection, the university had 3,714 degree-seeking graduate students. Of this population, 65.1% were master's students and 34.9% were doctoral students. Females comprised 57.9% of the graduate student body. Almost half (47.8%) were in the 25 to 34 years age range, but there were substantial numbers in the age ranges of less than 24 years of age (17.8%), 35 to 44 years (20.8%), and 45 years and older (13.6%). The university had 61.6% of its degree-seeking graduate students attending on a part-time basis (defined as less than 9 credit hours). Online graduate students made up 41.7% of graduate students. While we recognize that online surveys frequently have low response rates, the characteristics of our survey respondents (discussed next) show that we were able to capture a diverse student population.[3]

### Background of survey respondents

Our study sample leaned toward full-time students, on-campus students, and females. However, the purpose of our study was not to provide a representative analysis of graduate students at this particular university but to obtain a diverse sample of graduate students. Of the 343 degree-seeking graduate students comprising the study sample, 52.6% were master's students and 47.4% were doctoral students. As high as 70.3% of respondents were full-time graduate students and 32.1% were taking classes online. More than two-thirds (68.9%) of the sample was female. Almost half (45.3%) of graduate students in the sample were in the 26 to 35 years age range, while 16.1% were 25 years and under, 21.9% were in the 35 to 44 years category and 16.7% were 46 years and older. Graduate students participating in the survey relied on a wide range of sources of financial support. Over a third of respondents (35.0%) were employed full- time, while 36.7% relied on part-time employment on-campus as graduate assistants or equivalent. These graduate students also represented a wide range of academic disciplines, as shown in Figure 1.

### Operationalization and analysis

Our analysis approaches well-being in a holistic way, reflecting three dimensions: quality of life, physical health, and mental health. Quality of life was measured by the survey question "Overall, how would you describe your quality of life as a graduate student at [university]?" and physical health was measured using the question "Overall, how would you describe your physical health?" Responses ranged on a five-point scale from Poor to Excellent. We measured graduate students' mental health using the reported frequency of graduate students' feeling so depressed it was difficult to function, very sad, overwhelmed by all they had to do, and

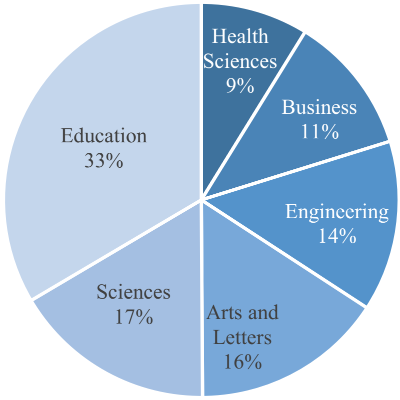


**Figure 1.** Breakdown of survey participants by academic college.

disconnected from friends and family. Specifically, the survey question asked: "During the past year, how frequently, if ever, have you …" with three response categories of Never or rarely, Occasionally, and Often. From these four mental health indicators we conducted principal factor analysis to create a factor score reflecting graduate students' underlying mental health status.[4] Finally, we created a wholistic measure of overall well-being as a factor score from our three dimensions quality of life and physical and mental health.[5]

To assess the role of work-life balance in graduate student well-being, we used the survey question: "Which of the following factors have had the most positive impact on your quality of life as a [University] graduate student?" Respondents were offered a list of 16 factors and could identify up to five. These factors, in addition to work-life balance, included diversity of students, faculty, and staff; quality of academic advising; extra-curricular activities; career and professional development activities; accessibility of faculty; and academic planning. As such, this work-life variable simply measures whether work-life balance was one of the top five factors impacting a graduate student's quality of life. We believe that this operationalization allows for a more nuanced understanding of the role of work-life balance, as it more narrowly captures those who have balance between work and life, and who perceive that balance to positively impact their quality of life.

We consider program climate as a factor that may be related to graduate student well-being. Program climate was measured as a factor score reflecting agreement (on a 5-point scale) with five statements: the intellectual climate in my program is positive; the social climate in my program is positive, my academic program creates a collegial and supportive environment, students in my academic program are treated with respect by faculty, and my program's procedures are fair and equitable to all.[6] We similarly measured sense of belonging, another factor expected to be related to well-being, as the level of agreement with the statement: I feel a sense of belonging to the [university] community.

To identify key sources of stress, we asked graduate students to identify up to five sources of stress from a list of 28 stressors, including finances, family issues, faculty advisor, academic workload, and time management. Similarly, to identify key sources of support, the survey asked the question: "Where did you go for assistance or support in addressing obstacles you identified or in managing stress?" For this question, respondents could select multiple options from the 14 response options provided, as applicable. Response options included academic advisor, peers or friends, parent or other family member, university health services, graduate program director, university counseling services, and non-university mental health professional.

To better understand issues of well-being and work-life balance across different types of graduate students, we conducted comparisons based on program level (master's vs. doctoral), full-time status (full-time vs. part-time), modality (on-campus vs. on-line), and gender. For comparisons, we used crosstabulation, one-way ANOVA, and pairwise correlation. We relied on $p < .05$ as the threshold value for statistical significance.

## Results

### Graduate student well-being

Figure 2 presents summary results of graduate student well-being. One out of every four (25.1%) graduate students in our study sample reported poor or fair quality of life and approximately one in five (22.5%) graduate students had poor or fair physical health. In terms of frequency of experience with mental health problems, approximately one out of every ten (14.5%) were often so depressed it was difficult to function, two in ten were often very sad (20.5%) and felt disconnected from friends and family (23.5%), and five in ten (53.2%) were often overwhelmed by all they had to do.

### Quality of life

Overall, most students described their quality of life as excellent (8%), very good (26%), and good (41%). However, 21% rated their quality of life as fair and 4% as poor. Figure 3 summarizes perceptions of quality of life for master's and doctoral students. The differences between master's and doctoral students were not statistically significant ($\chi 2 = 6.30$, $p = .178$). We also did not find statistically significant differences in quality of life when comparing graduate students according to modality (on-campus vs. on-line). Quality of life was statistically significantly lower for full-time students compared to part-time students ($\chi^2_{(4\ \mathrm{df})} = 16.29$, $p = .003$).

### Physical health

In terms of physical health, 12.7% of graduate students indicated having excellent health, while 30.4% indicated their health as very good and 34.3% as good. In contrast, 17.3% and 5.2% reported their physical health as fair and poor, respectively. The differences in physical health were not statistically significant by program level ($\chi^2(4\ \mathrm{df}) = 4.95$, $p = .292$) or gender ($\chi^2(4\ \mathrm{df}) = 2.08$, $p = .722$). However, part-time graduate students were statistically significantly healthier than full-time students ($\chi^2(4\ \mathrm{df}) = 12.24$, $p = .016$). For example, 15.7% of part-time graduate students indicated their health was excellent while 11.2% of full-time students reported excellent health. On the opposite end of the health spectrum, 10.8% of part-time graduate students indicated their health was fair or poor, compared to 27.8% of full-time graduate students.

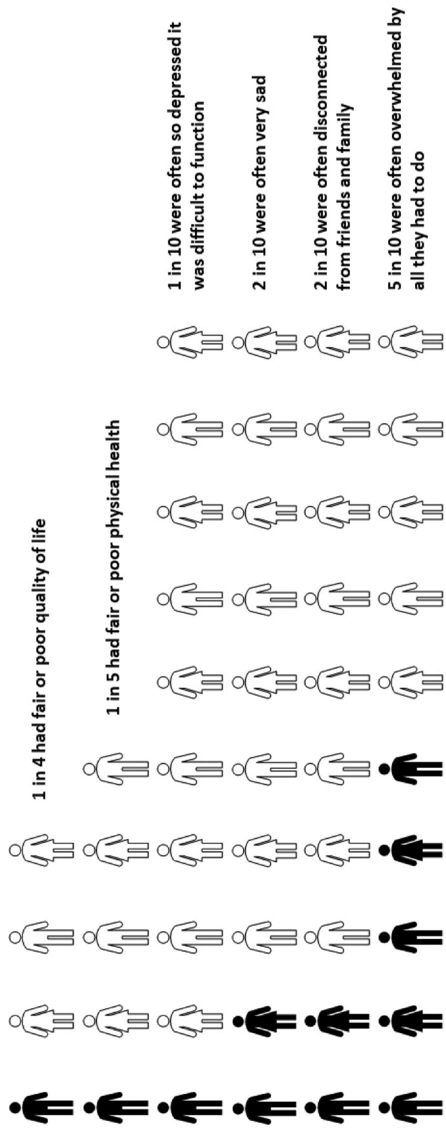


**Figure 2.** Graduate student well-being.

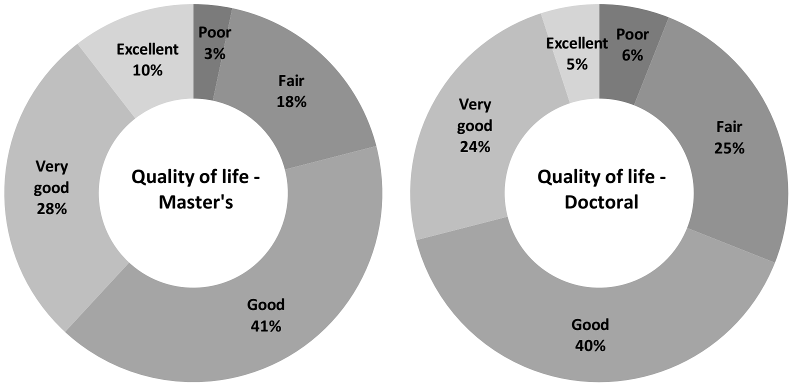


**Figure 3.** Perception of quality of life as a graduate student.

### *Mental health*

Figure 4 summarizes the frequency with which graduate students experienced mental health issues. Across all mental health indicators, master's students were not statistically different than doctoral students. For example, 50.8% of master's students and 55.5% of doctoral students reported often feeling overwhelmed by all they had to do ($\chi^2_{(2\ df)}$=1.69, p=.430).

We found no statistically significant differences in mental health status factor scores between master's and doctoral students, as tested via one-way ANOVA ($F_{(1df)}$=0.63, $p$=.429), or by gender ($F_{(1\ df)}$=1.81, $p$=.180) and full-time status ($F_{(1\ df)}$=0.63, $p$=.429). There were statistically significant differences between on-campus and on-line students ($F_{(1\ df)}$=8.86, $p$ = .003), with on-campus students having lower mental health scores.

### *Overall well-being*

We found that overall well-being does not vary by program level ($F_{(1\ df)}$=3.04, $p$=.082) or gender ($F_{(1\ df)}$=0.06, $p$=.802). Significant differences do exist, with lower well-being experienced by full-time graduate students ($F_{(1\ df)}$=12.95, $p$=.0004) and on-campus students ($F_{(1\ df)}$= 8.79, $p$=.003).

### *Work-life balance, well-being, and related factors*

Almost a third (32.7%) of graduate students indicated that work-life balance was one of the top five factors that positively impacted their quality of life. More master's students indicated this positive impact compared to doctoral students, but the difference was not statistically significant ($\chi^2_{(1\ df)}$=1.64, $p$=.201). The difference was also not significantly different by gender ($\chi^2_{(1\ df)}$= 2.33, $p$ = .127). However, having work-life balance that positively impacted quality of life was recognized by more part-time students (42.0%) compared to full-time students (29.0%), which was statistically significant ($\chi^2_{(1\ df)}$= 4.80, $p$=.029).

Furthermore, students whose work-life balance had a positive impact on their quality of life had better well-being. For example, only 11.5% of respondents who indicated work-life balance impacted their quality of life rated their quality of life as fair or poor, compared to 31.8% of students who did not identify a positive impact of work-life balance ($\chi^2_{(4\ df)}$= 30.49,

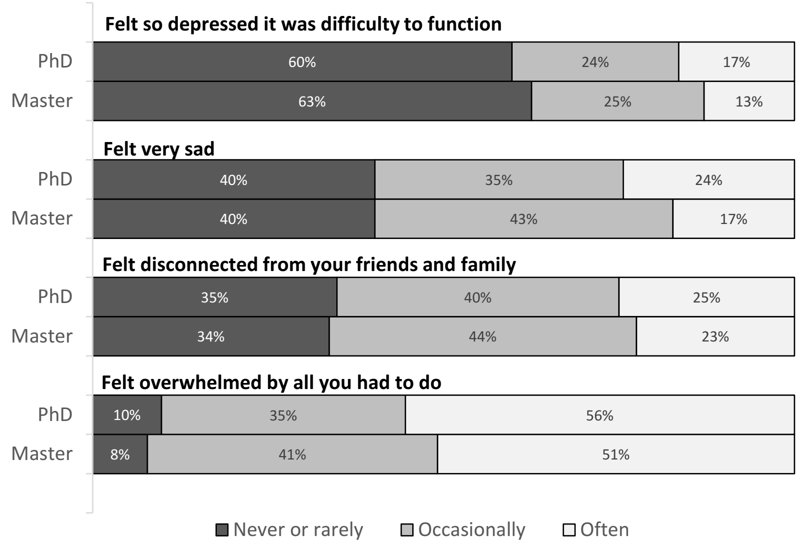


**Figure 4.** Frequency of experience with mental health issues.

$p$ = <0.0001). Furthermore, 8.0% of students whose work-life balance impacted their quality of life often felt depressed compared to 17.6% of those who did not ($\chi^2_{(4\ df)}$= 10.78, $p$=.005).

As shown in Table 1, four other factors were cited more frequently as having a positive effect on quality of life, three of which focus on relationships – with peers or fellow students, faculty, and faculty advisors or mentors. The quality of the academic program also contributed to well-being, and this issue may be related to the climate of the students' academic programs.

### *Program climate*

In terms of program climate, we found that 63.8% of graduate students agreed or strongly agreed that their program's social climate is positive. In contrast, 71.7% agreed that their academic program creates a collegial and supportive environment, and 80.3% agreed that their program's intellectual climate is positive. More than three-fourths (78.1%) of graduate students agreed that students in their academic programs are treated with respect by faculty and another three-fourths (73.8%) agreed that their program's procedures are fair and equitable. These results suggest that while graduate students value relationships with their peers, academic programs are doing a better job at creating a supportive and collegial intellectual climate, but less so at creating a supportive social environment.

**Table 1.** Top five factors that had the most positive impact on quality of life.

| | Percentage |
|---|---|
| Relationship with peers/fellow students | 56.1% |
| Quality of education/academic program | 45.7% |
| Relationship with faculty | 44.5% |
| Relationship with faculty advisor/mentor | 39.3% |
| Work-life balance | 32.7% |

Respondents could select up to five responses.
Respondents were not asked to rank-order the responses.

The program's overall climate was found to be positively related to well-being. Graduate students who perceived their program's climate more positively reported more positive overall well-being (r = 0.400, $p$ < .0001). Program climate was similarly related to work-life balance – graduate students whose work-life balance positively impacted their quality of life had more positive assessment of program climate ($F_{(1\ df)}$=5.19, $p$=.023). However, overall program climate did not significantly vary by program level ($F_{(1\ df)}$=3.36, $p$=.068), full-time status ($F_{(1\ df)}$=3.29, $p$=.070), or modality ($F_{(1\ df)}$=3.03, $p$=.083).

### *Sense of belonging*

Overall, 18.8% of respondents disagreed that they felt a sense of belonging to the university community, 27.4% were neutral and 53.9% agreed. Sense of belonging was related to well-being (r = 0.2978, $p$ < .0001), as those who felt a sense of belonging had more positive overall well-being. However, as shown in Table 2, sense of belonging did not vary by program level ($\chi^2_{(4\ df)}$= 11.78, $p$=.779), full-time status ($\chi^2_{(4\ df)}$2.505, $p$=.644), or modality ($\chi^2_{(4\ df)}$= 0.81, $p$=.937).

### *Sources of stress*

Given concerns about graduate student mental health, we examined sources of stress for graduate students. As shown in Table 3, outside responsibilities appear to be key stressors, including work obligations, finances, and home/domestic responsibilities. Coupled with time management, course requirements, and academic workload as top stressors, the results paint the picture of graduate students who face stress in their academic and personal lives.

We organized the sources of stress into three categories: "work" sources of stress related to school and academic life such as academic workload and course requirements; "life" sources of stress related to non-academic life such as work obligations and social life; and stressors resulting from balancing work and life.

In terms of sources of stress related to "work", there were some notable differences between master's and doctoral students as well as full-time and part-time students (see Table 4). For example, course requirements as a stressor was more prominent among master's students than among doctoral students ($\chi^2_{(1\ df)}$= 8.80, $p$=.003). In contrast, research was a stressor for more doctoral students than master's students ($\chi^2_{(1\ df)}$= 22.88, $p$=0<0.0001). Similarly, more full-time students identified research as a source of stress compared to part-time students ($\chi^2_{(1\ df)}$= 12.14, $p$<.0001). Faculty advisors ($\chi^2_{(1\ df)}$= 5.20, $p$=.023) and interactions with faculty and department ($\chi^2_{(1\ df)}$ = 5.53, $p$=.019) were stressors for doctoral students more so than for master's students. Course scheduling was a stressor

**Table 2.** Agreement with "I feel a sense of belonging to the [university] community".

| | Disagree | Neutral | Agree |
|---|---|---|---|
| **Program Level** | | | |
| Master's program (n = 176) | 18.8% | 26.7% | 54.5% |
| Doctoral program (n = 164) | 18.9% | 28.0% | 53.0% |
| **Full-time Status** | | | |
| Full-time (n = 238) | 17.7% | 26.5% | 55.9% |
| Part-time (n = 102) | 21.6% | 29.4% | 49.0% |
| **Modality** | | | |
| On-campus (n = 230) | 19.1% | 26.1% | 54.8% |
| On-line (n = 110) | 18.2% | 30.0% | 51.8% |

**Table 3.** Sources of stress.

| Work schedule/obligations | 57.2% |
|---|---|
| Finances | 54.9% |
| Time management | 40.5% |
| Future employment and career prospects | 32.4% |
| Course requirements | 31.2% |
| Academic workload | 28.9% |
| Home or domestic responsibilities | 24.6% |
| Physical and/or mental health | 20.8% |
| Research requirements | 19.4% |
| Family issues | 16.5% |
| Isolation/loneliness | 15.3% |
| Academic program structure or requirements | 14.7% |
| Course scheduling and/or availability | 12.1% |
| Interactions with faculty/department | 11.0% |
| Faculty advisor | 9.5% |
| Dating or social life | 8.7% |
| Childcare and/or eldercare | 5.5% |

Respondents could select up to five responses.
Respondents were not asked to rank-order the responses.

**Table 4.** "Work" sources of stress by program level and full-time status.

| | Master's students (n = 176) | Doctoral students (n = 164) | Part-time (n = 102) | Full-time (n = 238) |
|---|---|---|---|---|
| Course requirements | 38.0% | 23.2% | 29.4% | 31.5% |
| Academic workload | 31.8% | 26.2% | 29.4% | 29.1% |
| Research | 9.5% | 29.9% | 7.8% | 24.0% |
| Academic program structure or requirements | 14.0% | 15.9% | 11.8% | 16.2% |
| Course scheduling and/or availability | 16.2% | 7.9% | 9.8% | 13.3% |
| Interactions with faculty/department | 7.3% | 15.2% | 10.8% | 11.2% |
| Faculty advisor | 6.2% | 13.4% | 4.9% | 11.6% |

that differentially impacted master's students more than doctoral students ($\chi^2_{(1\ df)} = 5.54$, $p$=.020). These results are not surprising given differences in how programs are structured, how students interact with faculty and the department, and research expectations. However, our findings provide empirical evidence of the disparate academic stressors faced by graduate students at different program levels.

Life sources of stress varied mostly between part-time and full-time students. Specifically, as shown in Table 5, more part-time students were stressed because of their work schedule compared to full-time students ($\chi^2_{(1\ df)}$= 22.14, $p$<.0001). In contrast, more full-time students indicated finances as a stressor compared to part-time students ($\chi^2_{(1\ df)}$= 8.40, $p$=.004). Home or domestic obligation was a source of stress for more part-time students than full-time students ($\chi^2_{(1\ df)}$= 5.70, $p$=.017). Dating or social life, while a stressor for fewer graduate students, was a stressor for a larger proportion of master's students than doctoral students ($\chi^2_{(1\ df)}$= 7.90, $p$=.005).

In terms of stress associated with work-life balance issues (see Table 6), time management was a top stressor for graduate students, irrespective of program level, modality, and full-time status. Isolation and loneliness as a source of stress also did not significantly vary across program level, modality, and full-time status. However, more on-line students experienced stress from their physical and mental health compared to on-campus students, and the difference was significant ($\chi^2_{(1\ df)}$= 6.27, $p$=.012).

**Table 5.** "Life" sources of stress by program level, gender, and full-time status.

| | Master's students (n = 176) | Doctoral students (n = 164) | Female (n = 230) | Male (n = 104) | Part- time (n = 102) | Full- time (n = 238) |
|---|---|---|---|---|---|---|
| Work schedule/ obligations | 60.3% | 53.7% | 61.3% | 50.0% | 76.5% | 49.0% |
| Finances | 56.4% | 53.7% | 57.4% | 51.0% | 43.1% | 60.2% |
| Future employment and career prospects | 32.4% | 32.9% | 33.0% | 31.7% | 26.5% | 35.3% |
| Home or domestic responsibilities | 28.5% | 20.7% | 26.5% | 22.1% | 33.3% | 21.2% |
| Family issues | 15.1% | 18.3% | 17.8% | 15.4% | 22.6% | 14.1% |
| Dating or social life | 12.9% | 4.3% | 9.1% | 7.7% | 11.8% | 7.5% |
| Childcare and/or eldercare | 6.2% | 4.9% | 7.4% | 1.9% | 8.8% | 4.2% |

We examined how these stressors are correlated with graduate student well-being. Students who identified finances as a key stressor experienced lower levels of overall well-being ($r = -0.126$, $p = .020$) and more negative mental health ($r = -0.135$, $p = .012$). Academic workload was also negatively related to overall well-being ($r = -0.138$, $p = .011$) and mental health ($r = - 0.159$, $p = .003$). Not surprisingly, isolation/loneliness and physical and/or mental health were negatively (and significantly) correlated with all measures of well-being. Faculty advisor as a stressor, experienced by 9.5% of graduate students but disproportionately by doctoral and full-time students, was negatively correlated with overall well-being, quality of life, and mental health.

Given current concerns about graduate student mental health, we asked about support services used by graduate students in dealing with mental health issues and stress. Our results indicate very low utilization of professional support services – 9% of graduate students reported going to the university counseling services for support, 15% sought support from non-university mental health professionals, 3% relied on the university's health services, and 8% obtained services from non-university health professionals. These findings suggest that a university with a wide range of students especially in terms of local and on-line students – such as the one used for our study – must also rely on non-institutional resources to support graduate students.

### *Sources of support*

Our survey asked graduate students to indicate their sources for assistance or support in addressing obstacles or managing stress. Results indicate that students' support structure included a variety of sources for assistance. As shown in Figure 5, these sources, in decreasing order of reliance, were peers/friends; spouse, partner, or significant other; parent or other family member; academic advisor; other faculty; and graduate program director.

**Table 6.** Work-life balance sources of stress by program level, full-time status, and modality.

| | Master's students (n = 176) | Doctoral students (n = 164) | Part- time (n = 102) | Full-time (n = 238) | On- campus (n = 110) | On-line (n = 233) |
|---|---|---|---|---|---|---|
| Time management | 39.7% | 41.5% | 45.1% | 39.1% | 45.5% | 38.2% |
| Physical and/or mental health | 21.8% | 19.5% | 18.6% | 21.6% | 12.7% | 24.5% |
| Isolation/loneliness | 12.9% | 18.3% | 13.7% | 16.2% | 10.9% | 17.6% |

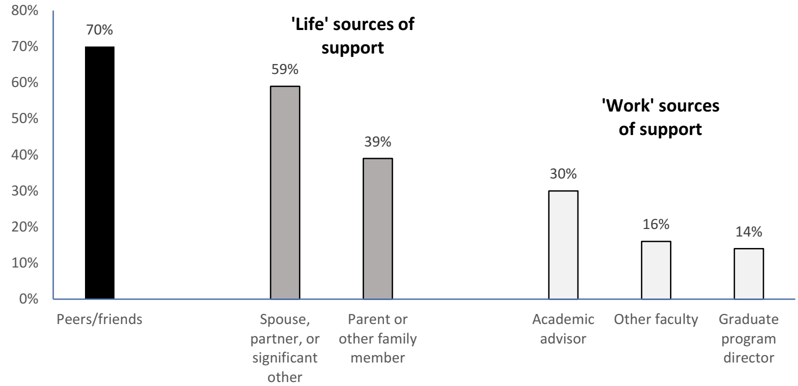


**Figure 5.** Sources of assistance or support.

While graduate students' support structure included a combination of sources from both their academic (i.e., work) and life domains, there was greater reliance on "life" sources of support compared to those from the work domain.

Sources of support varied across different groups of students, including master's and doctoral, male and female, and part-time and full-time students (see Table 7). In particular, more female students indicated reliance on peers and friends ($\chi^2_{(1\ df)}$= 10.37, $p$=.001) and on spouses, partners, or significant others ($\chi^2_{(1\ df)}$= 10.31, $p$=.001) as sources of support. More doctoral students compared to master's students ($\chi^2_{(1\ df)}$= 13.73, $p$<.001) and full-time students compared to part-time students ($\chi^2_{(1\ df)}$= 9.82, $p$=.002) find support in their academic advisors.

## Discussion and conclusion

### *Key findings*

Most previous studies did not compare graduate students' work-life balance and well-being across different program types, full-time status, or learning modalities. Our study offers more nuanced findings of work-life balance, well-being, and related factors for a diverse group of graduate students. Overall, we find that graduate students were challenged in terms of well-being. Most graduate students reported good to excellent quality of life, but there is a concern for the sizable percentage who reported poor to fair quality of life. Their physical

**Table 7.** Sources of assistance or support by program level, gender, and full-time status.

| | Master (n = 176) | Doctoral (n = 164) | Female (n = 230) | Male (n = 104) | Part- time (n = 102) | Full- time (n = 238) |
|---|---|---|---|---|---|---|
| Peers/friends | 68.8% | 72.0% | 74.3% | 56.7% | 68.6% | 71.0% |
| Spouse, partner, or significant other | 59.7% | 57.9% | 63.9% | 45.2% | 64.7% | 56.3% |
| Parent or other family member | 43.8% | 34.1% | 40.0% | 35.6% | 36.3% | 40.3% |
| Academic advisor | 22.2% | 40.2% | 30.0% | 29.8% | 18.6% | 36.1% |
| Other faculty | 14.8% | 17.1% | 14.8% | 19.2% | 14.7% | 16.4% |
| Graduate program director | 13.1% | 15.9% | 13.9% | 15.4% | 14.7% | 14.3% |

health primarily ranged from good to excellent. In terms of mental health, the majority of graduate students reported often feeling overwhelmed by all they must do. Similar to other studies (UC Berkeley Graduate Assembly, 2014; Cornwall et al., 2019; Juniper et al., 2012), multiple sources of stress, such as finances, academic workload, and isolation or loneliness are negatively related to well-being.

We found that graduate students whose work-life balance positively impacted their quality of life also had higher overall well-being. Relationships with peers, faculty, and advisors also positively impacted quality of life. The academic program's overall climate, rated favorably by most graduate students in our study, was also positively related to self-reported overall well-being. About half of the students reported feeling a sense of belonging to the university community, which was also related to well-being.

We found no substantial differences between master's and doctoral students across most factors with the exception of stressors and sources of support. Not surprisingly, course requirements and course scheduling were stressors for master's students while research, faculty advisor, and interactions with faculty and department were stressors for doctoral students. Academic advisors were also a source of support for more doctoral students than master's students.

Full-time student status was a much stronger differentiator compared to program level. Fewer full-time students, compared to part-time students, perceived work-life balance as positively impacting their quality of life. Full-time and part-time students also varied in their sources of stress and sources of support. Surprisingly, there is little evidence of significant differences across genders, except for reliance on different sources of support. This finding runs contrary to the studies that found that more women reported lack of work-life balance and greater mental health issues than men (Barreira et al., 2018; Evans et al., 2018; Haynes et al., 2012; Kulp, 2020). Although more on-campus students had lower mental health and overall well-being than on-line students, more on-line students experienced stress from physical and mental health than on-campus students.

## Implications and recommendations

This study provides insights into work-life balance and well-being across a diverse mix of graduate students. Our results clearly identify that work-life balance is a key factor contributing to graduate students' well-being and that several work and life factors are key stressors. Our findings point to the need to understand the differences between types of graduate students in a more nuanced way beyond simple comparison by program level, such as undergraduate vs. graduate or master's vs. doctoral. Rather, we find that the paths – through which work and life introduce stress, provide sources of support, and affect well-being – can vary depending on program level, full-time student status, program modality, and gender.

For example, part-time graduate students experienced positive impacts of work-life balance on quality of life to a greater extent than full-time students. Efforts to help graduate students manage stress must first recognize that the various populations have different stressors and acknowledge that university support programs and infrastructure may only address some facets of the work-life equation. However, by recognizing the differences across sub-populations of graduate students, universities can target specific groups with support they need the most.

Furthermore, our results suggest that university-related sources of support – faculty, advisors, and graduate program directors – are relied on to a much lesser degree than personal or life-related sources. At the same time, relationships with faculty and advisors contribute to graduate students' quality of life. Aside from programmatic changes to address sources of stress, building stronger relationships within programs by fostering a culture where faculty are more accessible to and supportive of graduate students can have positive implications.

University administrators, faculty, advisors, and staff can use our results to enhance graduate student well-being by developing a supportive infrastructure that emphasizes both the social and academic climates and builds a sense of community and belonging. Our results also suggest that universities need to continue providing adequate mental health services and facilitate graduate student access to such services.

Universities need to also recognize that changes by both institutions and individual students are necessary to address work-life imbalances among graduate students, the stress resulting from balancing work and life, and impacts on well-being. Thus, in addition to institutional infrastructure and support, universities should consider strategies that empower graduate students to manage stress, build a support network, and become part of the department and university community. Students should be provided the infrastructure, programmatic support, and personal opportunities to balance the demands of work and life and to enhance their well-being.

### *Limitations*

There are some limitations to this study. First, while our study is inclusive of a wide range of graduate students, our study context is a research-intensive public university in the U.S. Thus, our findings may not be applicable to graduate students at universities with a dissimilar research and educational focus. However, our findings are consistent with other studies that have found similar concerns. Furthermore, given our inclusive sample of diverse graduate students, our findings provide a baseline understanding of key aspects of the graduate student experience that may be relevant to other universities in the U.S.

Second, our survey had a relatively low response rate. However, our sample represented a diverse group of graduate students. The range of responses to our survey questions further suggests the lack of voluntary response bias, where responses can be skewed by respondents with strong opinions. Future research can enhance our study methodology by improving the response rate, using larger sample sizes, and/or extending the study to multiple universities.

Third, we offer an important caveat. Our use of self-reported assessments of well-being may understate concerns about graduate student well-being, particularly in terms of their actual mental health. Given the stigma associated with mental health issues, fewer graduate students may be willing to accurately report their mental health experiences. Barreira et al. (2018) found that most graduate students in their study tended to overestimate the quality of their mental health, so students in our sample may have poorer mental health than we report in our findings. Future research could utilize a combination of well-being measures to overcome this concern.

Lastly, we are not evaluating possible causal factors in our analysis of relationships between key constructs. It is possible that graduate students' perception of their own well-being may color how they perceive their connection to academic programs (i.e., sense of belonging), assess the impact of their work-life balance, respond to potential stressors, and perceive factors such as program climate and the availability of social support. More theory-based research is needed to provide better understanding of these causal relationships.

### Future research and practice

Our study's empirical findings clearly identify that work-life balance is critical to graduate student well-being and that several work and life factors are key stressors. However, our study does not consider how actors – graduate students, faculty, advisors, department chairs, deans, and others – participate in transforming or sustaining the structures that enable or constrain graduate student efforts to manage work and family demands. For future research, we turn to the three theories introduced earlier as starting points for identifying how policies and programs can help students manage different roles and responsibilities in their work and life domains, and how institutional practices can be more responsive to work-life balance and well-being issues of graduate students.

Social cognitive theory posits that students, in managing work and life responsibilities, will pursue actions that they believe can be successful and lead to desirable outcomes, within the existing program or university structure, for ensuring work-life balance. As such, consideration of graduate students' perceived self-efficacy is important when developing organizational policies and programs that emphasize work-life balance and structures that empower graduate students. Addressing work-life balance and well-being issues, then, require addressing graduate students' perceived self-efficacy to balance work and life responsibilities, while also creating a university climate where students' actions can meaningfully impact the organizational culture to make it more conducive for work-life balance and well-being. University programs and climates that empower graduate student may help enhance graduate students' self-efficacy, but institutional actors also need to consider how socio-cultural factors reduce self-efficacy, exacerbate work-life imbalance, and negatively impact well-being.

The concept of duality from structuration theory has implications for changing institutional culture and climate to better support graduate student work-life balance and well-being. Culture and climate are systems that are perpetually negotiated (and re-negotiated) and produced (and re-produced) through interactions that may alter the structure over time. People vary in the extent to which they behave in ways consistent with, or contrasted to, extant structures indicating variation in both capacity for agency and the extent to which agency is exercised to create change (Sewell, 1992). Our empirical findings show that graduate students exhibited variation in their connections to other actors within the organization (peers, faculty, advisors, etc.), their perceptions of the academic and social climate, and their sense of belonging to the university. These factors can contribute to different agency among graduate students to "buck the system" and behave in ways conducive to work-life balance. Other actors, such as faculty and advisors, may possess more power and agency to create change. Therefore, structuration theory is useful for understanding how actors embedded within institutions and the institutions themselves – academic programs, departments, and/or the university – interact to create, adopt, and implement rules, and create resources that support graduate student work-life balance and enhance well-being.

Structuration theory further highlights the importance of organizational policies that affect graduate student work-life balance. Every social institution or organization relies on formal and informal policies to coordinate participants' activities. These policies, as mediating resources, can be used to create a uniform, routinized expectation for action (Canary & Tarin, 2017). However, such policies also serve as regulatory mechanisms for governing graduate student behavior consistent with specific expectations of those with power. As such, while work-life friendly policies may enable activities in support of work-life balance, other policies can simultaneously constrain those activities. Furthermore, resistance to work-life friendly policies or informal meanings that are inconsistent with the formal policy can shape how policies are implemented in practice. It is important to examine the confluence of structural elements and discourse that shape not only how policies regarding work-life balance are developed and implemented but also how members of the organization interpret and respond to policies over time.

Finally, work/family border theory offers a framework for how balance between work and life may be attained (Clark, 2000; Schieman et al., 2009). The interface between work and life is also socially constructed (Clark, 2000; Kreiner et al., 2009; Nippert-Eng, 2008), with graduate students as active agents in the co-construction of boundaries through continuous interaction with others. The theory, then, is useful for determining how graduate students manage boundaries and engage with others in negotiating demands between work and life domains. Kreiner et al. (2009) identified four types of boundary work tactics that could be utilized to create appropriate work-life segmentation or integration. Training for graduate students in managing work-life balance could include how these tactics can be used in different settings.

Border theory offers utility in understanding the roles of faculty, advisors, and peers. These border-keepers are influential in defining the domain and border, given their power over the graduate student border-crosser. They play an important role in the graduate students' ability to manage borders and domains. However, they can also be barriers if their definitions of what constitutes work and life are based on their own limited experience or if they are too rigid in guarding the domains and borders to the extent that it limits border-crossers' abilities to deal with conflicting demands. Academic programs and universities should encourage high life-domain awareness among faculty and advisors so that they better recognize graduate students' non-work commitments, responsibilities, and needs.

Members of the work domain can also be encouraged to increase their commitment to the border-crosser. Such commitment "is manifested by caring about the border-crosser as a total person, not just in terms of how the border-crosser fills one's immediate needs" (Clark, 2000, p. 763) and can lead to increases in the well-being (Clark, 2000; Galinsky & Stein, 1990; Greenhaus, 1988; Repetti, 1987). Universities and individuals within universities need to consider how they can show more commitment to graduate students navigating work-life balance issues and how such commitment can increase graduate students' well-being.

## Notes

1. Our research was approved as exempt human subjects research. To ensure anonymity of research participants, our survey instrument could not include questions that identified the race of the respondent or whether the respondent was a domestic or international student.
2. These include the UC Graduate Student Well-Being Survey Report (https://www.ucop.edu/institutional-research-academic-planning/_files/graduate_well_being_survey_report.pdf), the MIT Enrolled Graduate Student Survey (http://ir.mit.edu/graduate-enrolled-student-surveys), the University of Washington College of Education Annual Graduate Student Survey Report (https://www.education.uw.edu/coeir/past-surveys/), and the Temple University Graduate Student Survey

(http://www.temple.edu/ira/documents/assessment/student-surveys/Graduate-Student-Survey-Report.pdf).

3. We assessed non-response bias by dividing the sample into three response waves of approximately equal numbers and compared our key variables of interest – quality of life, physical health, mental health, and work-life balance – between the first and third wave. We found the differences were not statistically significant.
4. The Bartlett test of sphericity was statistically significant ($p<0.0001$) and the Kaiser-Meyer-Olkin measure of sampling adequacy was 0.765. One factor with an eigenvalue of 2.24 was extracted. All variable loaded onto this one factor with factor loadings over 0.56.
5. The Bartlett test of sphericity was statistically significant ($p<0.0001$) and the Kaiser-Meyer-Olkin measure of sampling adequacy was 0.677. One factor with an eigenvalue of 1.20 was extracted. All variable loaded onto this one factor with factor loadings over 0.60.

   Descriptive statistics for variables used in our analysis are not reported in this article but can be obtained by contacting the authors.
6.. The Bartlett test of sphericity was statistically significant ($p<0.0001$) and the Kaiser-Meyer-Olkin measure of sampling adequacy was 0.845. One factor with an eigenvalue of 2.92 was extracted. All variable loaded onto this one factor with factor loadings over 0.69.

## Notes on contributors

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